

EEG-Based Same-Limb Movement Classification: Overcoming CSP Limitations with Deep Learning

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Abstract

Classifying electroencephalogram (EEG) signals for motor imagery or execution is a foundational task in Human-Machine Interfaces (HMI). While the Common Spatial Pattern (CSP) algorithm coupled with Linear Discriminant Analysis (LDA) is highly effective for binary inter-limb tasks (e.g., left vs. right hand), its performance degrades significantly when classifying different movements originating from the same limb. This paper investigates the limitations of the traditional CSP-LDA pipeline for intra-limb movement classification and proposes three deep learning architectures—EEGNet, EEGPsdNet, and EEGConnNet—to overcome these shortcomings. By analyzing latent space representations and classification outcomes, we demonstrate the necessity and efficacy of deep neural networks in extracting complex spatial-spectral features for high-resolution EEG movement classification.

Index Terms—EEG motor decoding; common spatial patterns; deep learning; multimodal fusion; upper-limb classification.

1. Introduction

The ability to accurately decode complex upper-limb movement states from non-invasive scalp electroencephalography (EEG) is a critical driver for advancements in brain-computer interfaces (BCIs), targeted neurorehabilitation, and robotic assistive devices [1]. However, signal quality at the scalp is inherently limited by spatial mixing due to volume conduction, low signal-to-noise ratio (SNR), and high inter-trial variability. These factors make multiclass intra-limb decoding particularly challenging compared to standard binary inter-limb tasks.

To address these challenges, this study systematically compares two distinct analytical frameworks applied to identical cue-locked data. **Block I** serves as the baseline, utilizing a classical *CSP+LDA pipeline*: handcrafted spatial filters optimized for discriminative variance, followed by a linear Gaussian classifier [2]. **Block II** introduces a set of *convolutional decoders*. This includes a standard EEGNet-style architecture [3] operating on filtered, cue-locked voltage epochs, alongside two custom dual-branch models. These custom models are designed to include auxiliary features derived from

Welch power spectral density (PSD) maps and pairwise wavelet cross-spectral connectivity.

This comparative approach demonstrates that transitioning from traditional linear models to deep learning networks with temporal awareness significantly improves classification accuracy. Furthermore, we show additional performance gains by integrating multimodal feature fusion.

Data. All experiments utilize the open-source Graz movement-execution corpus [1]. Code and preprocessing configurations are publicly available in the GitHub repository referenced in the Acknowledgments.

Figure 1 illustrates the condition-averaged sensorimotor waveforms obtained after the initial filtering and segmentation pipeline.

2. Materials and Methods

2.1. Dataset and Preprocessing Pipeline

This study utilized the openly available Graz movement-execution EEG corpus [1], recorded while volunteers performed sustained upper-limb actions including elbow, forearm, and hand synergies, alongside a resting state

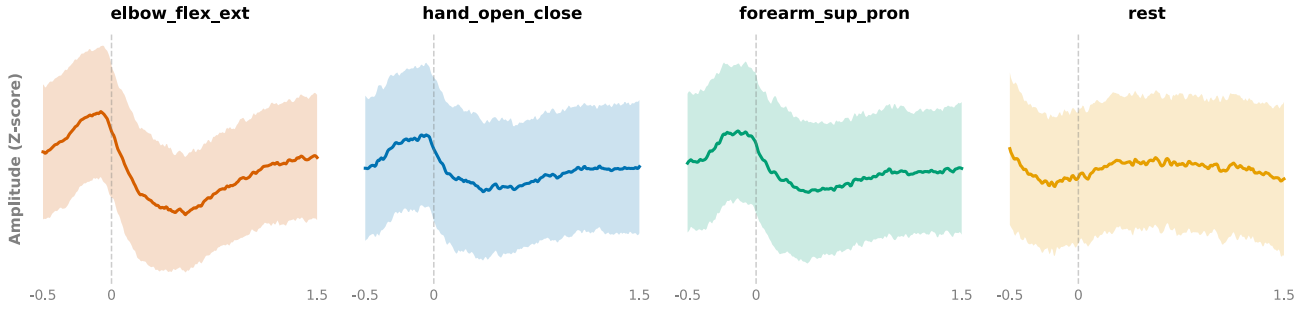


Figure 1 Grand-average cue-locked EEG waveforms.

(BNCI Horizon 2020, accession 001–2017). To focus on the sensorimotor cortex, analyses were restricted to twenty-one peri-central electrodes spanning the fronto-central, central, and centro-parietal regions (FC5–FC6, FCz, C5–C6, Cz, CP5–CP6, CPz). The spatial coverage of this selected motor montage is illustrated in Fig. 2.

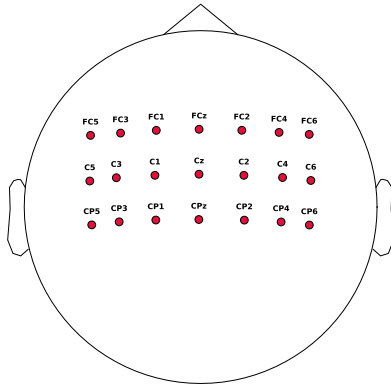


Figure 2 Topographical layout of the 21 selected peri-central EEG channels used for feature extraction and classification, targeting the sensorimotor cortex.

Continuous EEG signals were downsampled to 128 Hz and subjected to a mains-notch filter, followed by an FIR band-pass filter between 0.5 and 40 Hz using zero-phase kernels to preserve temporal dynamics. Cue-locked epochs were extracted from -0.5 to 2.0 s relative to the stimulus onset. To ensure signal quality, trials containing peak EEG magnitudes exceeding $\pm 200 \mu\text{V}$ were discarded prior to any data augmentation. Finally, mirror movement directions (e.g., flexion/extension, supination/pronation, open/close) were pooled to form a robust four-class intra-limb classification problem: *elbow*, *hand*, *forearm*, and *rest*.

2.2. Feature Extraction and Multimodal Tensors

To evaluate the efficacy of multimodal data fusion in deep learning architectures, three distinct representa-

tional tensors were extracted from the preprocessed epochs (as illustrated in Fig. 3):

- **Temporal:** The core input consisted of per-trial, per-electrode temporally normalized EEG voltage epochs.
- **Spectral:** Welch power spectral density (PSD) maps were computed between 3 and 30 Hz, capturing the logarithmic-magnitude frequency responses associated with motor execution.
- **Connectivity:** Phase-coupling descriptors were derived from the modulus of pairwise complex Morlet cross-spectral densities at 10 Hz (μ -band) and 22 Hz (β -band). These were summed over unordered electrode pairs and globally standardized to represent functional network connectivity.

2.3. Classification Architectures

The classification task was approached using two distinct paradigms: a classical spatial filtering baseline (Block I) and a set of differentiable temporal receptive field networks (Block II).

2.3.1 Block I: CSP with LDA Baseline

The baseline implemented the canonical motor-BCI blueprint [2]. Common Spatial Pattern (CSP) filters were optimized to maximize the logarithmic-band-power variance ratios between classes. The first six orthogonal spatial patterns were selected to project the EEG epochs into a reduced feature vector ($\mathbb{R}^6$), which was subsequently classified using Linear Discriminant Analysis (LDA).

2.3.2 Block II: Convolutional Neural Networks

Three convolutional architectures were evaluated to assess the impact of deep feature extraction and multimodal fusion:

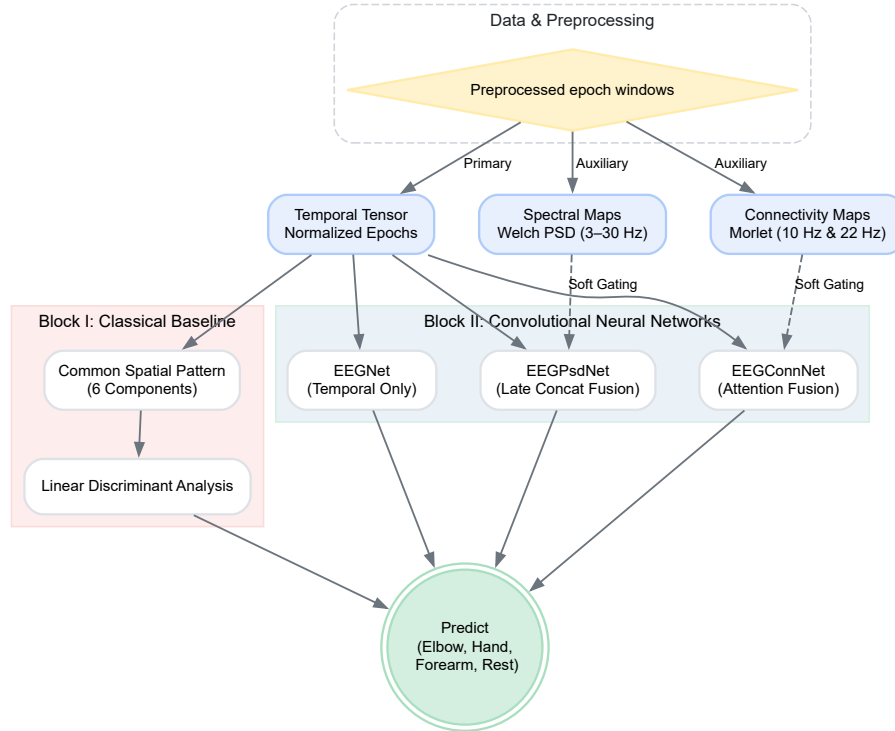


Figure 3 Schematic of the preprocessing and feature extraction pipeline feeding the proposed deep learning architectures. The PSD and connectivity branches serve as optional inputs depending on the specific model variant evaluated.

- **EEGNet Baseline:** A compact, separable convolutional network designed directly for filtered EEG epoch tensors [3].
- **EEGPsdNet (Spectral Fusion):** A dual-branch architecture employing late fusion. Embeddings from the temporal CNN stem were concatenated with shallow convolutional embeddings extracted from the Welch PSD maps. The fused vector was batch-normalized prior to the final softmax classification layer.
- **EEGConnNet (Connectivity Fusion):** An independent encoder architecture that processed waveform embeddings alongside the spatial coherence tensors. Softmax attention mechanisms were utilized to dynamically modulate modality weights before passing the features through a Multi-Layer Perceptron (MLP) for classification.

Figure 4 provides a visual schematic of the three evaluated convolutional paradigms. While the baseline EEGNet relies purely on spatio-temporal filtering of the time-domain epochs, EEGPsdNet and EEGConnNet introduce distinct late-fusion mechanisms to integrate spectral and spatial connectivity priors, respectively.

2.4. Training and Evaluation Protocol

All trainable deep architectures were evaluated using a stratified random partition of the pooled subject data, allocating approximately 85% of trials for training and 15% for validation. A fixed pseudo-random seed was maintained across all runs to ensure comparable convergence noise.

Network optimization was performed by minimizing a weighted cross-entropy loss function with mild label smoothing. Models were trained using the AdamW optimizer with weight decay. To ensure stable convergence, learning rates were dynamically reduced upon reaching a plateau in validation loss, and gradients were clipped. Additive Gaussian noise was applied to the EEG tensors as a data augmentation strategy to prevent overfitting. Checkpoints yielding the maximum validation accuracy were retained for final inference.

To interpret the learned feature spaces, withheld softmax precursor vectors were extracted and projected into two dimensions using *t*-Distributed Stochastic Neighbor Embedding (*t*-SNE) with matched hyperparameters across all Block II models.

3. Experimental Results

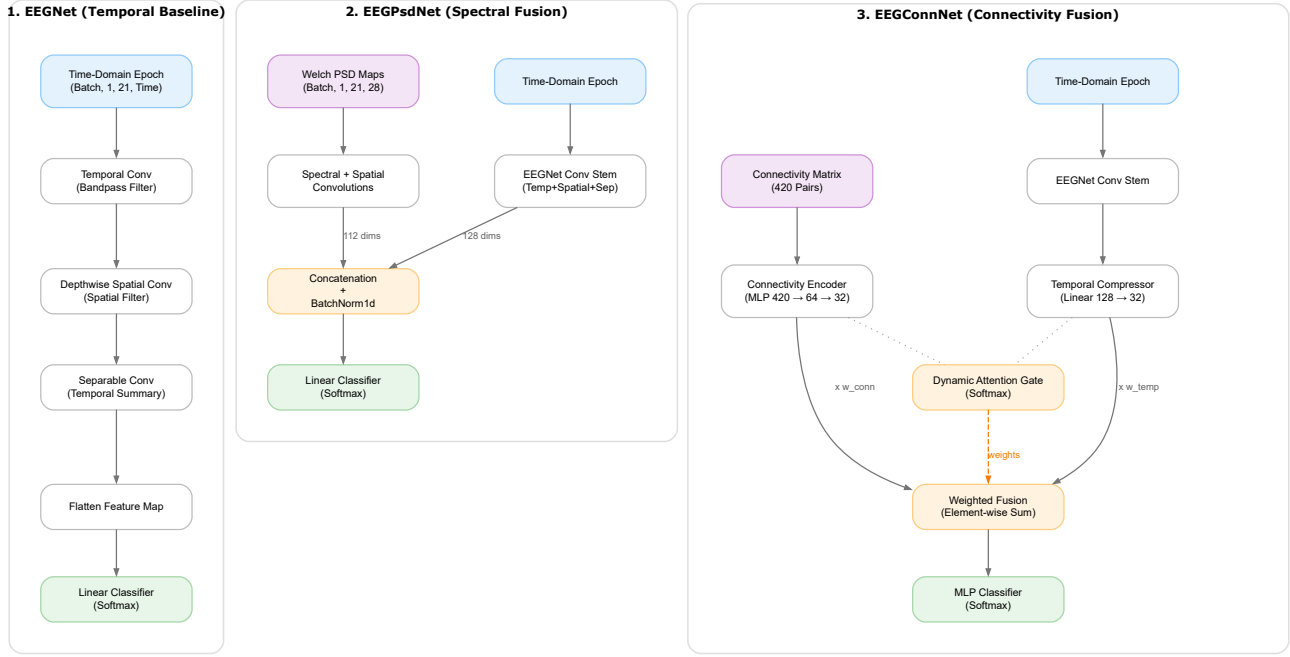


Figure 4 Schematic overview of the three evaluated convolutional architectures. The baseline EEGNet utilizes strictly temporal and spatial convolutions. EEGPsdNet implements a late-fusion concatenation with spectral maps, while EEGConnNet employs a dynamic attention gate to adaptively weight temporal and connectivity embeddings prior to the final classification.

3.1. Latent Representations and Error Dynamics

To understand the performance disparities between the classical pipeline and the proposed deep learning architectures, we directly compared their pre-decision latent spaces alongside their resulting confusion matrices on the withheld validation set. This comparative 4×2 grid is presented in Figure 5.

The classical CSP+LDA baseline (Fig. 5a, b) highlights the fundamental difficulty of intra-limb classification. While CSP spatial filters effectively map the *rest* state, the feature distributions for active synergistic movements (*elbow*, *hand*, *forearm*) are heavily overlapped in the primary component space. Consequently, the linear classifier suffers from severe off-diagonal confusion, failing to establish robust boundaries between active states.

Transitioning to the convolutional baseline, EEGNet (Fig. 5c, d) demonstrates the capacity of temporal receptive fields to begin untangling these manifolds. The *t*-SNE projection reveals distinct cluster formation, significantly improving global accuracy. However, noticeable bridging remains between the *hand* and *forearm* geometries, resulting in a misclassification bottleneck.

The integration of multimodal priors actively resolves these boundary ambiguities. Both EEGPsdNet (Fig. 5e, f) and EEGConnNet (Fig. 5g, h) force tighter intra-class clustering and wider inter-class margins. EEGConnNet,

driven by cross-spectral attention, exhibits the most isolated latent representations, successfully minimizing synergistic confusion and consolidating the diagonal accuracy matrix.

3.2. Overall Performance Benchmarks

The quantitative impact of these feature space improvements is summarized in Table 1. Models were evaluated based on maximum validation accuracy during optimization, with the classical baseline fitted purely on the training partition to ensure an unbiased comparison.

The baseline CSP approach achieved approximately 36% accuracy, reinforcing its limitations for same-limb paradigms. The baseline temporal CNN elevated this to ~56%. However, the dual-branch architectures incorporating late-fusion spectral embeddings (EEGPsdNet) and dynamic cross-spectral attention (EEGConnNet) achieved the highest generalization, both scoring between 61% and 62%. This confirms that supplementing raw voltage dynamics with structural network connectivity yields superior decoding bounds for complex motor imagery.

3.3. Neurophysiological Validation

To ensure the highest performing connectivity-based model was grounding its predictions in physiological

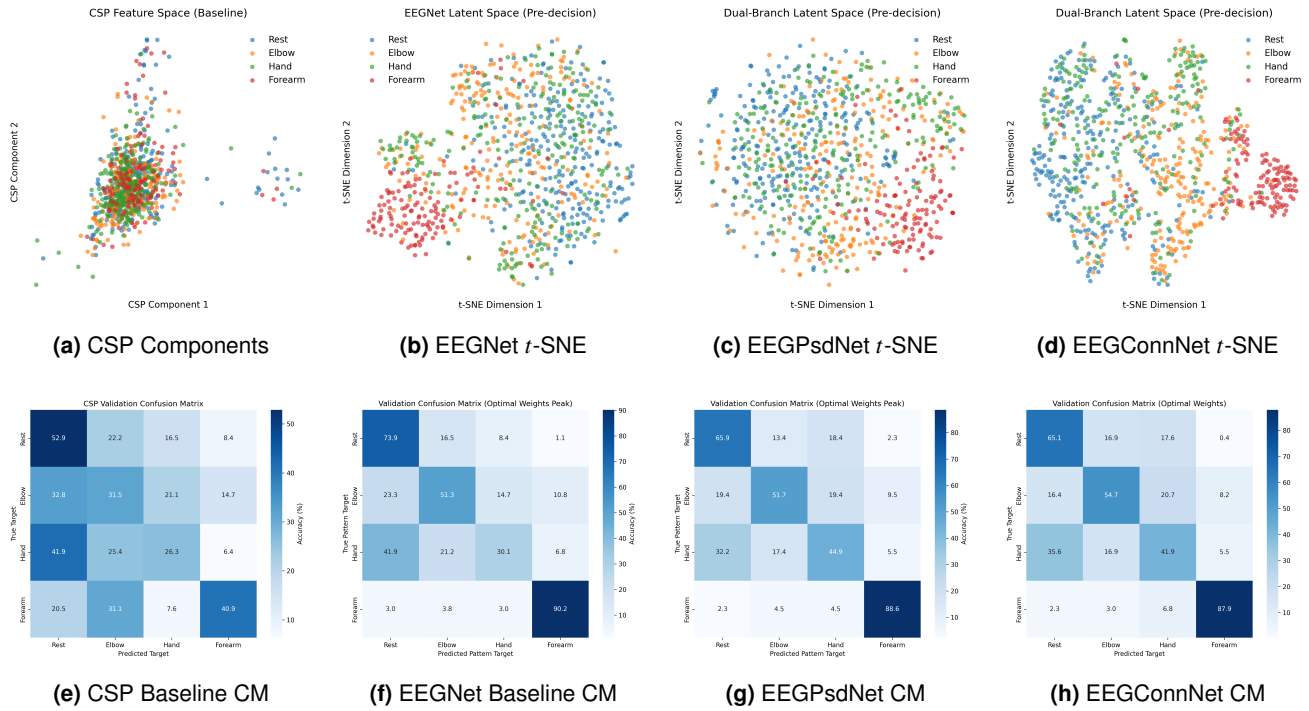


Figure 5 Comprehensive comparison of pre-decision feature geometries (top row) and resulting classification error distributions (bottom row) across the evaluated frameworks. Moving from left to right illustrates how the sequential integration of deep temporal filters and multimodal priors progressively disentangles the complex intra-limb overlap observed in the classical spatial baseline.

reality rather than overfitting to noise or artifacts, we projected its primary feature weights back into the electrode space. Figure 6 summarizes the trial-averaged connectivity-derived spatial patterns contrasting hand synergy epochs against the resting state. The resulting topology confirms that the network successfully isolated task-relevant nodes localized over the contralateral sensorimotor cortex, validating the neurophysiological integrity of the fusion network's decision mechanism.

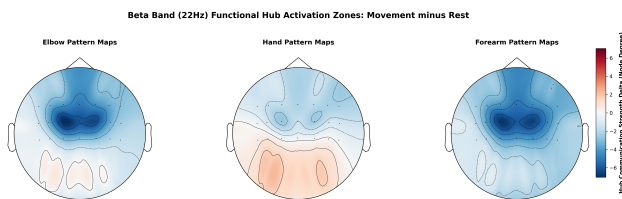


Figure 6 Trial-averaged connectivity-derived spatial patterns contrasting hand synergy epochs versus rest on the standard Graz layout. Activation is appropriately localized to the sensorimotor areas.

4. Discussion

Breaking the Limits of Linear Spatial Filtering. The classification of intra-limb movements presents a severe challenge due to the heavily overlapping cortical activation patterns within the same hemisphere. As

demonstrated in Table 1, the classical exploratory CSP pipeline hovered near chance-level accuracy ($\sim 36\%$) for this four-class task. The transition to convolutional receptive fields (Block II) yielded an immediate improvement of approximately 20 percentage points over the optimized shallow CSP baseline. This substantial leap suggests that successful same-limb decoding relies heavily on non-linear spatio-temporal dynamics that simple spatial variance filters inherently fail to capture. However, it is worth noting that for complete parity in future benchmarking, CSP and CNN pipelines must strictly unify band-pass envelopes, referencing, and temporal cropping before blaming spatial filtering deficits alone.

The Efficacy and Boundaries of Multimodal Fusion. While deep temporal filters began untangling the latent manifolds, synergistic movements (e.g., hand versus forearm) still suffered from significant misclassification bottlenecks. Integrating multimodal priors successfully addressed this. Welch PSD concatenation and cross-spectral coherence attention each added roughly 5 percentage points of accuracy over the EEGNet-style baseline. The t -SNE embeddings in Figure 5 visually confirm that fusion forces tighter intra-class clustering and resolves boundary ambiguities. Nevertheless, because withheld softmax inputs from different classes

Table 1 Held-out four-class classification accuracy, parameter footprint, and architectural notes evaluated on the withheld validation fraction.

Model Name	Description & Representations	Model Size	Accuracy [%]
Block I: CSP+LDA [2]	Motor-band CSP features after grid search across band hypotheses; classified via standard LDA.	10KB	~36
Block II: EEGNet [3]	Temporal CNN baseline operating directly on filtered, cue-locked voltage epochs using separable convolutions.	15KB	~56
Block II: EEGPsdNet	Late fusion architecture combining temporal waveform embeddings with Welch PSD maps (3–30 Hz).	39KB	~61
Block II: EEGConnNet	Dual-branch encoder processing raw waveforms and cross-spectral coherence matrices via dynamic modality attention.	160KB	~61 – 62

remain neighboring points across all Block II models we conclude that auxiliary maps supply marginal, complementary constraints rather than entirely orthogonal, linearly separable geometry

Neurophysiological Validity versus Raw Accuracy. In neurotechnology, interpretable connectivity structures and final softmax margins are not interchangeable metrics While Table 1 shows only modest incremental accuracy gains from connectivity fusion Figure 6 proves that the network’s attention mechanisms are physically grounded. The organized peri-central contrast between trial-averaged hand epochs and the resting state confirms that the fusion network relies on genuine sensorimotor structures rather than overfitting to hardware noise or muscular artifacts

Generalization and Future Work. While this study establishes the superiority of deep multimodal fusion for complex upper-limb decoding, the current validation strategy relies on withholding trials from a pooled subject mix This approach is inherently optimistic relative to the strict leave-one-subject-out (LOSO) cross-validation required for clinical neurotechnology translation Recommended follow-ups include strictly subject-independent benchmarks to evaluate true model generalization alongside the exploration of fusion designs that exploit auxiliary tensors without weakening the underlying temporal discriminability

Acknowledgment

Code:

<https://github.com/MNR-MarcosOriolPagonabarraga-projects/HMI-EEGMovementClassification.git>

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